



An experimental and modeling study of methyl propanoate pyrolysis at low pressure



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ABSTRACT

Methyl propanoate (MP) pyrolysis in a laminar flow reactor was studied at low pressure (30 Torr) within the temperature range from 1000 to 1500 K. About 30 products were detected and identified in the pyrolysis process using the photoionization mass spectrometry, including H₂, CO, CO₂, CH₃OH, CH₂O, CH₂CO, C1 to C4 hydrocarbons and radicals (such as CH₃, C₂H₅ and C₃H₃). Their mole fraction profiles versus temperature were also measured. For the unimolecular dissociation reactions, the rate constants were calculated by high precision theoretical calculations. Based on the theoretical calculations and measured mole fraction profiles of pyrolysis species, a kinetic model of MP pyrolysis containing 98 species and 493 reactions was developed. The model simulates the primary decomposition process well with the calculated rate constants. According to the rate of production analysis, the decomposition pathways of MP and the formation channels of both oxygenated and hydrocarbon products were discussed. It is concluded that the main decomposition pathway is $\text{MP} \rightarrow \text{CH}_2\text{COOCH}_3 \rightarrow \text{CH}_3\text{CO} + \text{CH}_2\text{O} \rightarrow \text{CO}$.

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1. Introduction

As a potential substitute of petroleum-derived diesel fuels, bio-diesel attracts increasing public concerns due to many advantages such as the renewability, low CO₂ emission and low sulfur content. The increasing concerns on the combustion chemistry of long-chain fatty acid methyl esters (FAMES) which are dominant components of bio-diesel require thorough investigations on the special reaction kinetics induced by the ester group. Combustion studies of short-chain fatty acid methyl esters can isolate the ester structure from the long-chain alkane structures, and facilitate research on combustion characteristics of FAMES [1]. As a representative short-chain fatty acid methyl ester, methyl propanoate (MP) is the smallest methyl ester containing a unique C–C bond alpha to the carbonyl group in the molecular structure. Besides, MP is an intermediate with relatively high concentrations in the pyrolysis of bio-diesel such as the rapeseed methyl ester (RME) [2]. Therefore, understanding of the reaction kinetics of MP will contribute to the development of a reliable kinetic model for larger methyl esters [1].

Combustion research of MP can be traced back to 1960s. Research interests were focused on comparing the cool flame and

ignition characteristics between MP and other oxygenates with analogous molecular structures. Salooja investigated the ignition of MP and other oxygenated hydrocarbons in 1966 [3]. No cool flame was observed for MP in his work. Hoare et al. also studied the oxidation characteristics of MP, they obtained similar results to Salooja's [4,5]. Recently, Farooq et al. investigated the pyrolysis of methyl acetate, MP and methyl butanoate in shock tubes [6]. They measured the CO₂ yield over temperatures from 1260 to 1653 K, pressures from 1.4 to 1.7 atm, and reactant concentrations from 2% to 3%. Akih-Kumgeh and Berghthorson studied the impact of alkyl chain length on fuel reactivity for C1–C4 alkanolic acid methyl esters using shock tube [7]. They also presented the computational results for methanol elimination from MP using the CBS-QB3 method. There are some kinetic models for the pyrolysis and combustion of esters. However, previous experimental investigations on MP combustion provided insufficient identification and concentration measurements of combustion intermediates for the validation of kinetic models. The sub-mechanism of MP is still not complete [8–10], which limits the understanding of MP chemistry.

In this work, the pyrolysis of MP has been studied in a laminar flow reactor at 30 Torr over the temperature range of 1000–1500 K using synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS). Low-pressure condition can enhance the sensitivity of primary decomposition reactions of MP to its consumption, reduce secondary reactions, and are helpful for the

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detection of radicals which are important intermediates for validating kinetic mechanisms. The utilization of SVUV-PIMS can facilitate the comprehensive identification of pyrolysis products including both stable molecules and active radicals, and provide mole fraction measurements of these species. Based on the experimental results and theoretical calculations on elementary reactions in MP sub-mechanism, a detailed kinetic model of MP pyrolysis has been developed and validated.

2. Experimental section

The experiments were performed at the National Synchrotron Radiation Laboratory in Hefei, China. The detailed description of the two beamlines used in this work can be found elsewhere [11–14]. Briefly, the VUV light from the bending magnet beamline covers the photon energy ranging from 6.5 to 11.7 eV with a LiF window to eliminate the higher-order harmonic radiation, while the undulator beamline covers the photon energy ranging from 8.0 to 17.0 eV and uses a gas filter filled with Ar or Ne to eliminate higher-order harmonic radiation. Both beamlines use 1 m Seya–Namioka monochromator with energy resolving powers of 500 and 1000, respectively.

The pyrolysis apparatus has been reported in detail previously [15–17]. The top panel of Fig. 1 shows the schematic diagram of the pyrolysis apparatus. It consists of a pyrolysis chamber, a differentially pumped chamber with a molecular-beam sampling system, and a photoionization chamber with a home-made reflectron time-of-flight mass spectrometer (RTOF-MS) with mass resolving power of ~ 2000 . The pyrolysis chamber contains a laminar flow reactor, which is used for maintaining the low pressure circumstance. The flow reactor is mainly composed of a high temperature furnace and a flow tube which is made of α -alumina and has a 150 mm heating length in the furnace. In previous laminar flow reactor studies, Cathonnet et al. [18] and Glarborg et al. [19,20] suggested using very thin inner diameters for flow tubes (8 mm in [18] and 9 mm in [19,20]) to ensure strong radial

diffusion effects, reduce radial concentration gradients, and achieve adequately homogenous reaction circumstances. Therefore the inner diameter of our flow reactor was selected to be 6.8 mm which is much thinner than those in previous studies [18–20]. Furthermore, the wall catalytic effects are negligible in the experiments because of the weak catalytic effects of α -alumina [21] used in our flow tube and the low specific surface area inside the polished flow tube.

MP was vaporized at 373 K and mixed well with the diluted gas Ar in a vaporizer. The mixture was then transported along a long stainless steel tube (also heated to 373 K) before entering the reaction region in the flow reactor. The pyrolysis species were sampled at the outlet of the flow tube by a quartz nozzle with 40° cone angle and a ~ 500 μm orifice at the tip, forming a molecular beam in the differentially pumped chamber. They finally passed into the photoionization chamber through a nickel skimmer. In the ionization region, the molecular beam was crossed and ionized by the tunable synchrotron VUV light. The produced ions were detected by the RTOF-MS.

In this work, MP (99%) was purchased from Sinopharm Chemical Reagent Co., Ltd. in Shanghai, China; Ar (99.999%) was purchased from Nanjing Special Gases Ltd. in China. The gas flow rate of Ar was controlled at 980 standard cubic millimeters per minute (SCCM) by a mass flow controller. The liquid flow rate of MP was controlled at 0.086 ml/min (equivalent to 20 SCCM in gas phase at room temperature) by a liquid chromatography pump. Thus, the total flow rate is 1000 SCCM with the initial MP mole fraction of 2.0%. In this work, the pressure at the outlet of the flow tube was kept at 30 Torr.

The temperature distribution inside the flow tube was measured using a Pt–10%Rh/Pt thermocouple for the simulation of MP pyrolysis. The measurement method was reported in detail previously [16], thus only a brief introduction is provided herein. A W–Re thermocouple was put close to the middle region of the heating wire to monitor the outside temperature of the flow tube, which is named T_{out} . Before performing the experiment, another thermocouple (B-type) was put inside the flow tube to measure the temperature profiles along the flow tube. The uncertainty of the temperature measurement is within ± 30 K. As shown in Fig. 1 (the bottom panel), each temperature profile along the flow tube is named by its maximum temperature (T_{max}). As mentioned above, the pressure at the outlet of the flow tube was kept at 30 Torr. The Reynolds number (Re) of the flow was calculated to be less than 154 over the temperature range from $T_{\text{max}} = 830$ to 1670 K. Thus it is a laminar flow in view of $\text{Re} < 2000$. For the pressure distribution along the flow tube, the detailed information is provided in the Supplemental materials.

The experiment was carried out in two modes. (1) Mass spectra were recorded at a fixed pyrolysis temperature and various photon energies to yield photoionization efficiency (PIE) spectra. Species identification can be performed according to the measured molecular weight and ionization energy (IE). (2) Mass spectra were recorded at several fixed photon energies and various temperatures to yield mole fraction profiles of species versus temperature. In this work, the mass spectra were measured at photon energies of 16.64, 11.70, 11.00, 10.50, 10.00 and 9.70 eV to obtain near-threshold photoionization. The evaluation method of mole fraction was reported in detail in our previous work [11–13]. Briefly, for species i , the signal S_i can be expressed by,

$$S_i = C \times X_i(T, E) \times \sigma_i(E) \times D_i \times \Phi_p(E) \times \lambda(T) \quad (\text{E1})$$

where C is the proportion constant, $X_i(T, E)$ is the mole fraction of species i at the selected temperature (T) and photon energy (E), $\sigma_i(E)$ refers to the photoionization cross section which is only decided by photon energy, D_i is the mass discrimination factor, and $\Phi_p(E)$ is the photon flux at the photon energy E . The mole

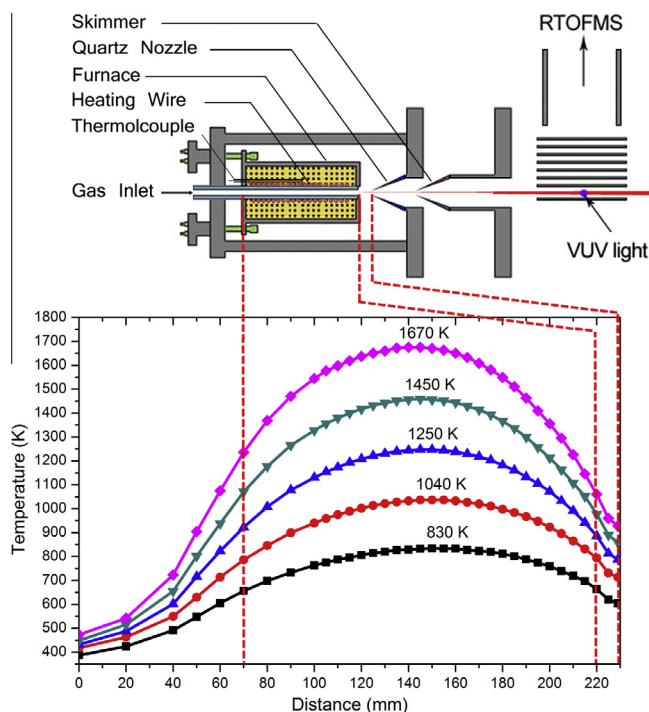


Fig. 1. Schematic diagram of the pyrolysis apparatus and measured temperature profiles along the reactor axis with five T_{max} s.

fractions of every species can be calculated at several photon energies as long as the selected energies exceed the ionization energy of the species. Generally, the calculation results at lower energy were chosen to avoid fragmentation interference at high photon energy. The photoionization cross sections (PICSSs) used in this work are from our online database [22] and previous work [23–25]. The experimental uncertainties of mole fractions are evaluated to be $\pm 25\%$ for the products with known PICSSs and a factor of 2 for those with estimated PICSSs. The experimental uncertainties of C, H, and O balances are all within $\pm 10\%$ compared with inlet fluxes of C, H, and O elements.

3. Theoretical calculations and kinetic model

3.1. Theoretical calculations

The major unimolecular decomposition pathways of MP were computed at the CBS-APNO level [26]. The geometry optimization and frequency calculation were performed at HF/6-311G(d,p) for CBS-APNO followed by a second geometry optimization at QCISD/6-311G(d,p) level. The further single point energy corrections including QCISD(T), MP2 and CBS extrapolation calculations provide the complete basis set extrapolation energy. All ab initio calculations were performed with the Gaussian 09 program [27].

The potential energy profiles of dominant decomposition pathways obtained at CBS-APNO are shown in Fig. 2, including direct bond cleavage and small molecular elimination reactions. Among them, three main reaction channels are:



As shown in Fig. 2, two reaction pathways forming CH_3CHCO and CH_3OH were found in this work, i.e. a one step pathway via a four-member-ring transition state (TS1) with the barrier height of 73.02 kcal/mol and a three-step pathway including H-migration, H-rotation and CH_3OH -elimination with the overall barrier of 75.21 kcal/mol (TS2). This is different from the results deduced by Akih-Kumgeh and Bergthorson [7]. Thus the final rate constant of R3 is the summation of the two pathways. The pressure-

and temperature-dependent rate constants of the channels R1–R3 were computed with the RRKM/master equation method by using the Variflex program [28] at the pressures of 5, 30, 70, 200 and 760 Torr with the temperature ranging from 800 to 2000 K. The rate constants were evaluated at the E/J (energy E and total angular momentum J) resolved calculation [29]. J ranges from 1 to 241 with a step size of 10 and E covers the range from -29918.04 to 50081.96 cm^{-1} with an energy grain size of 100 cm^{-1} , which is consistent with our previous work [30–32]. The pressure-dependent rate constants were calculated with one-dimensional (1D) master equation, while the master equation was solved by an eigenvalue-solver-based approach for the dissociation processes. The bond fission channels without apparent transition states (channels R1 and R2) were treated variationally in the temperature range by the procedure proposed by da Silva and Bozzelli [33]. The potential energy curves of the C–C and C–O bond dissociation of MP were calculated by the CAS-PT2(2e,2o)/6-311G(d,p)//CASSCF(2e,2o)/6-311G(d,p) method [34], scanning along the dissociation bond length with the interval of 0.2 Å. The active space was chosen as the bonding orbital σ and anti-bonding orbital σ^* for the C–C or C–O bond breaking. The low-frequency vibrational modes corresponding to internal rotation around the breaking bonds were assumed as free rotors. Other torsional modes were dealt with the hindered rotor approximation based on the hindrance potentials computed at B3LYP/6-31g(d) level. The channel R3 with tight transition states was treated with the rigid-rotor harmonic-oscillator assumption [35]. And the quantum tunneling effect was corrected according to the Eckart model [36]. The optimized geometries of the stationary points involved in MP pyrolysis at the CBS-APNO level were illustrated in Fig 3. The computational results of fundamental quantities including entropies, enthalpies of formation, and heat capacities at 1 atm are tabulated in Table 1 for the major species.

In this work, Ar was chosen as the bath gas. The Lennard–Jones (L–J) parameters of Ar employed in this calculation are $\sigma = 3.465 \text{ Å}$ and $\epsilon/k = 113.5 \text{ K}$. The L–J parameters for MP were computed with the following empirical equations [37]:

$$\sigma = 2.44 \times (T_c/P_c)^{1/3} \quad (\text{E2})$$

$$\epsilon_A/k_b = 0.77 \times T_c \quad (\text{E3})$$

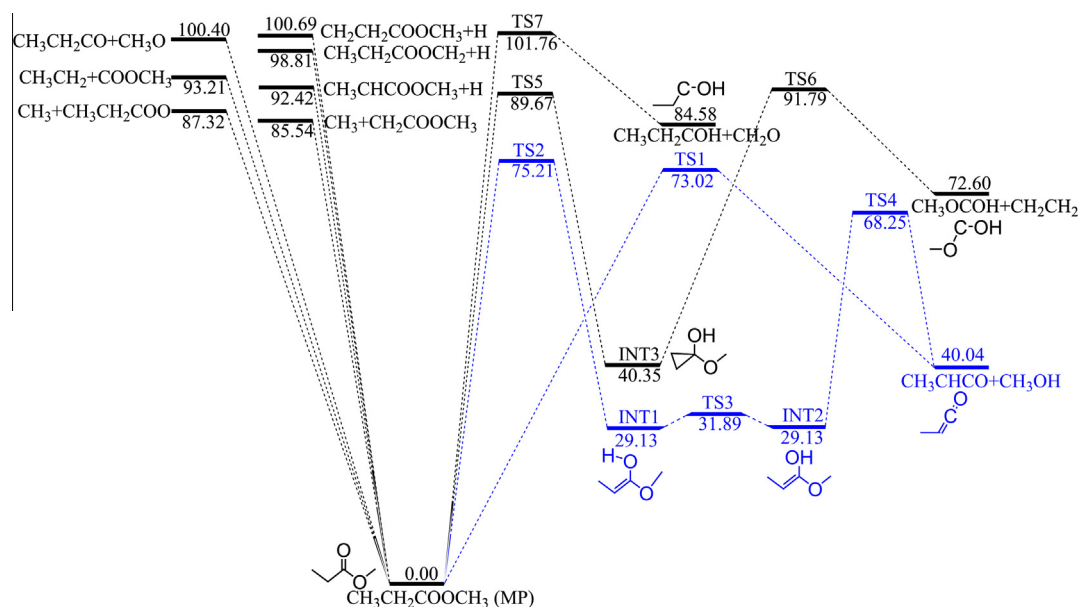


Fig. 2. The calculated major decomposition pathways of MP at the CBS-APNO level. The energies (in kcal/mol) with zero-point-energy correction are relative to MP.

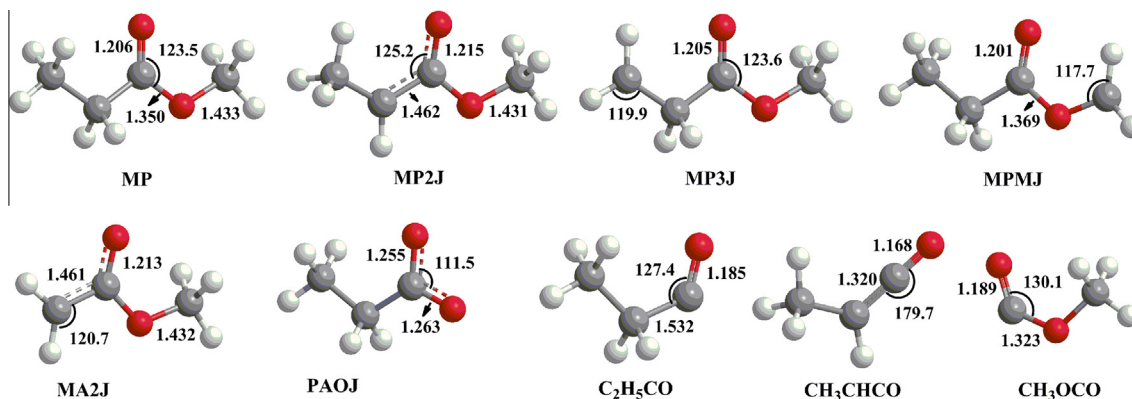


Fig. 3. The geometries of the stationary points involved in MP pyrolysis optimized at the B3LYP/6-31G(d) level.

Table 1

Calculated results of entropies, enthalpies at $T = 298.15$ K and C_p s at $P = 1$ atm.

Species	$S_{298.15}$	$H_{298.15}$	$C_{p298.15}$	C_{p500}	C_{p1000}	C_{p1500}	C_{p2000}	C_{p2500}	C_{p3000}
MP	82.5	−105.9	24.8	36.3	55.9	65.6	70.7	73.5	75.2
MP2J	84.7	−64.0	25.1	35.6	52.9	61.4	65.8	68.3	69.7
MP3J	87.8	−55.5	26.2	36.5	53.2	61.5	65.9	68.3	69.8
MPMJ	83.3	−57.7	25.3	36.2	53.2	61.5	65.9	68.3	69.8
MA2J	74.8	−53.2	20.4	28.4	40.8	46.9	50.0	51.8	52.9
PAOJ	76.8	−51.6	18.9	26.9	40.1	46.6	49.9	51.7	52.8
C_2H_5CO	70.8	−9.0	16.7	23.6	35.2	41.2	44.3	46.0	47.0
CH_3CHCO	67.4	−16.1	15.7	21.8	31.6	36.6	39.2	40.6	41.5
CH_3OCO	68.5	−39.4	14.6	19.4	27.8	32.0	34.1	35.3	36.0

Note: Units of S in $\text{cal mol}^{-1} \text{K}^{-1}$, H in kcal mol^{-1} , C_p in $\text{cal mol}^{-1} \text{K}^{-1}$.

where k_b is the Boltzmann constant. The critical temperature (T_c) and pressure (P_c) were cited from literature [38]. The calculated L–J parameters of MP are $\sigma = 5.776$ Å and $\varepsilon_A/k_b = 408.56$ K. The collision energy transfer was treated using an exponential-down model with $\langle \Delta E_{\text{down}} \rangle = 200(T/300)^{0.85} \text{ cm}^{-1}$. The computed rate constants are listed in Table 2. It is worth noting that various input parameters such as barrier heights, Lennard–Jones parameters and the collisional model used may introduce uncertainties to the calculated rate constant. In the present study, the uncertainty in the barrier heights is assumed to be approximately 1 kcal/mol with the CBS-APNO method [39]. The empirical equations for estimating Lennard–Jones parameters have been generally accepted, while the required parameters i.e. the critical temperature T_c and pressure P_c were accurately measured [38]. The temperature-dependent form of $\langle \Delta E_{\text{down}} \rangle$ used in the exponential-down model of collision energy transfer has been validated in previous studies [40,41], which should be more reasonable than the traditional choice by assuming ΔE_{down} as a constant. Given those uncertainties in the input parameters, we estimate the cumulative uncertainty in the calculated rate constants to be a factor of 2.

3.2. Kinetic model

The simulation work was performed using the Plug Flow Reactor (PFR) Code in Chemkin-PRO software [42]. The detailed simulation method can be found in our recent studies of *n*-butanol and 2-butanol [17,43]. Our previous work [17,43] also illustrated the compatibility of our pyrolysis models in simulating both our flow reactor data and shock tube pyrolysis data (e.g., those from Hanson's group [44]). Moreover, general decomposition processes of various fuels detected by our flow reactor experiments are also reproducible by the kinetic models developed by other research groups [45–50] in butanol studies. The measured temperature profiles and the calculated pressure profiles were used as input

parameters in the simulation. Detailed description of the simulation method can be found in our previous work [16,17] and also listed in the Supplemental materials.

The reaction mechanism of MP pyrolysis mainly consists of the sub-mechanisms of MP decomposition and C0–C4 chemistry. In this work, the development of C0–C4 sub-mechanism was based on the pyrolysis model of butene isomers [16], which was developed from the USC Mech II model [51] with updated C3 and C4 sub-mechanisms. In this work, the C3 and C4 sub-mechanisms were further updated with reactions taken from the studies by Kiefer et al. [52], Miller et al. [53] and Sarathy et al. [49]. The sub-mechanism of MP is proposed in this work, including unimolecular decomposition and H-abstraction reactions of MP, and the subsequent reactions of initial products. All the rate constants of unimolecular and H-abstraction reactions by the H radical and CH_3 group were newly calculated, as well as the subsequent reactions of the initial radical products, as listed in Table 2. The model includes 98 species and 493 reactions, which are shown in the Supplemental materials.

4. Results and discussion

A number of pyrolysis species including stable species and free radicals were identified in this work. Their molecular weights, formula, names, and measured IEs are presented in Table 3. The measurement of PIE spectra at 1400 K facilitates the identification of most isomers. For example, the peak at m/z 40 is distinguished as allene and propyne, m/z 42 as propene and ketene, and m/z 44 as carbon dioxide, aldehyde and propane. Products with relatively low signal intensities cannot be affirmed by PIE measurement.

The experimental (symbols) and simulated (lines) mole fraction profiles of MP, oxygenated products and hydrocarbon products are displayed in Figs. 4 and 5. Comparison between the experimental

Table 2
Rate constants calculated for reactions in MP pyrolysis model.

Order	Reactions	A	n	E
R1	MP = MA2J + CH ₃			
	0.006579 atm	3.530E+109	−27.34	135,663
	0.039474 atm	2.982E+102	−25.09	133,844
	0.092105 atm	1.826E+97	−23.51	131,563
	0.263158 atm	1.688E+89	−21.13	127,439
R2	MP = PAOJ + CH ₃			
	0.006579 atm	8.029E+102	−25.63	131,299
	0.039474 atm	3.403E+96	−23.54	130,492
	0.092105 atm	5.631E+91	−22.06	128,767
	0.263158 atm	1.053E+84	−19.74	125,154
R3	MP = CH ₃ OH + CH ₃ CHCO			
	0.006579 atm	7.320E+91	−22.68	118,841
	0.039474 atm	4.435E+84	−20.39	117,147
	0.092105 atm	1.443E+79	−18.73	114,695
	0.263158 atm	5.553E+70	−16.25	110,305
R4	MP = CH ₃ OCO + C ₂ H ₅			
	0.006579 atm	2.429E+101	−25.34	132,540
	0.039474 atm	3.499E+98	−24.19	134,810
	0.092105 atm	8.526E+94	−23.02	134,159
	0.263158 atm	2.655E+88	−21.02	131,742
R5	MP = C ₂ H ₅ CO + CH ₃ O			
	0.006579 atm	1.169E+77	−17.79	129,996
	0.039474 atm	1.829E+91	−22.06	143,638
	0.092105 atm	1.167E+90	−21.78	137,321
	0.263158 atm	2.073E+97	−24.03	148,982
R6	MP = MP3J + H	1.025E+13	−1.89	12,018
R7	MP = MP2J + H	8.283E+05	−0.19	10,639
R8	MP = MPMJ + H	1.530E−15	2.74	4457
R9	MP2J = MPMJ	3.760E+07	1.91	7585
R10	MP3J = MPMJ	4.035E+05	2.42	10,649
R11	MP + H = MP3J + H ₂	2.949E−01	3.90	14,427
R12	MP + H = MP2J + H ₂	5.752E+02	2.88	9941
R13	MP + H = MPMJ + H ₂	9.381E+00	3.41	13,367
R14	MP + CH ₃ = MP3J + CH ₄	8.100E+09	−0.55	2093
R15	MP + CH ₃ = MP2J + CH ₄	1.800E+15	−2.23	9006
R16	MP + CH ₃ = MPMJ + CH ₄	7.070E+66	−16.11	73,894
R17	PAOJ = CO ₂ + C ₂ H ₅	1.180E+45	−10.43	39,651
R18	C ₂ H ₅ CO = C ₂ H ₅ + CO	6.790E+44	−10.36	39,425
R19	MP2J = CH ₃ O + CH ₃ CHCO			
R20	MP3J = C ₂ H ₄ + CH ₃ OCO			
R21	MPMJ = CH ₂ O + C ₂ H ₅ CO			

Notes: (a) $k = AT^n \exp(-E/RT)$. The units are in cm³, mol, s, cal.

(b) The listed rate constants of R5–R21 were calculated in this work at the pressure of 30 Torr.

(c) MP: CH₃CH₂COOCH₃; MA2J: CH₂COOCH₃; PAOJ: CH₃CH₂COO; MP2J: CH₃CHCOOCH₃; MP3J: CH₂CH₂COOCH₃; MPMJ: CH₃CH₂COOCH₂.

and simulated results presents fairly good agreement, which indicates that the model can satisfactorily reproduce the thermal decomposition process of MP.

4.1. Decomposition pathways of MP

As shown in Fig. 4a, MP begins to decompose at around 1100 K. It decomposes quickly from 1200 to 1450 K, while most oxygenated intermediates have relatively high mole fractions within this temperature range. The decomposition pathways of MP are displayed in Fig. 6 based on the rate of production (ROP) analysis at 1400 K, where MP and most observed products have relatively high concentrations. Considering the importance of oxygen conversion in MP decomposition, Fig. 6 is plotted according to the reaction flux of O element among oxygenated species. The thickness of arrows denotes the flux of corresponding reactions.

The major consumption reactions of MP are unimolecular dissociation reactions (R1)–(R3). The C–C scission reaction producing CH₂COOCH₃ (MA2J) and methyl radicals (R1) consumes 47% of MP. MA2J has a resonantly stabilized structure; and it is more stable than PAOJ which is the product of R2. As shown in Fig. 2, the

Table 3
List of species detected in the pyrolysis of MP from 1000 to 1500 K.

m/z	Formula	Species	IE (eV)	
			Literature ^a	Experiment ^b
2	H ₂	Hydrogen	15.43	15.45
15	CH ₃	Methyl radical	9.84	9.84
16	CH ₄	Methane	12.61	12.62
18	H ₂ O	Water	12.62	12.65
26	C ₂ H ₂	Acetylene	11.40	11.40
28	C ₂ H ₄	Ethylene	10.51	10.53
	CO	Carbon monoxide	14.01	14.00
29	C ₂ H ₅	Ethyl radical	8.12	8.13
30	CH ₂ O	Formaldehyde	10.88	10.89
	C ₂ H ₆	Ethane	11.52	11.55
31	CH ₃ O	Methoxy radical	10.72	10.73
32	CH ₃ OH	Methanol	10.84	10.89
39	C ₃ H ₃	Propargyl radical	8.67	8.68
40	C ₃ H ₄	Allene	9.69	9.84
	Propyne		10.36	10.40
42	CH ₂ CO	Ketene	9.62	9.60
	C ₃ H ₆	Propene	9.73	9.75
44	CH ₃ CHO	Aldehyde	10.23	10.27
	C ₃ H ₈	Propane	10.94	10.95
	CO ₂	Carbon dioxide	13.78	13.80
52	C ₄ H ₄	Vinylacetylene	9.58	9.63
54	1,3-C ₄ H ₆	1,3-Butadiene	9.07	9.12
56	C ₃ H ₄ O	Methylketene	8.95	8.96
58	C ₃ H ₆ O	Propanal	9.96	10.00
72	C ₃ H ₄ O ₂	Propenoic acid	10.60	10.62
74	C ₃ H ₆ O ₂	Propanoic acid	10.44	10.46
86	C ₄ H ₆ O ₂	Methyl acrylate	9.90	10.04
88	C ₄ H ₈ O ₂	Methyl propanoate	10.15	10.14

^a Reference to NIST Chemistry Webbook [37].

^b The uncertainty of IE measurement is ±0.05 eV in this work.

dissociation energy of α-C–C bond is lower than that of the H₃C–O bond, resulting in high reaction flux of R1. Decomposition products of MP except PAOJ are finally converted to CO, as shown in Fig. 6. Among these pathways, decomposition reaction sequences of MA2J to both CH₃CO and CH₂O contribute almost half of CO yield.

According to the ROP analysis, MA2J is the most important precursor (63%) of CH₂O via a unimolecular dissociation reaction (MA2J = CH₃CO + CH₂O) and the reaction sequence of MA2J → MAMJ → CH₃CO + CH₂O. The appearance temperature of CH₂O (near 1150 K), which is just following the initial decomposition temperature of MP. It indicates that CH₂O is one of the most important initial products.

The unimolecular C–O scission reaction R2 to produce CH₃CH₂COO (PAOJ) and methyl radicals has a contribution of 14% for MP consumption. The formation of CO₂ almost completely relies on the unimolecular decomposition of PAOJ radical (R17, PAOJ = C₂H₅ + CO₂), which is produced by R2. According to the simulation result, the mole fraction of PAOJ is much lower than ppm level beyond the experimental detection limit. Given that PAOJ is an extremely reactive intermediate, the formed PAOJ would totally decompose via R17 immediately. Thus, it is concluded that R2 controls the reaction flux to CO₂ formation. This conclusion is verified by the sensitivity analysis which suggests that the formation of CO₂ is exclusively sensitive to R2 and shows no sensitivity to R17, as shown in Fig. 7. Therefore, the accuracy of the calculated rate constants of R2 are validated by the good agreement between the simulated and measured mole fraction profiles of CO₂. Pathways initiated with R1 and R2 lead to the formation of CO and CO₂, respectively. Great competition is presented between the processes producing CO and CO₂. This competition relationship is also confirmed by the sensitivity analysis in Fig. 7. CO and CO₂ are located at the end of the decomposition reaction sequences of MP, and are the destination of almost all oxygen elements in the

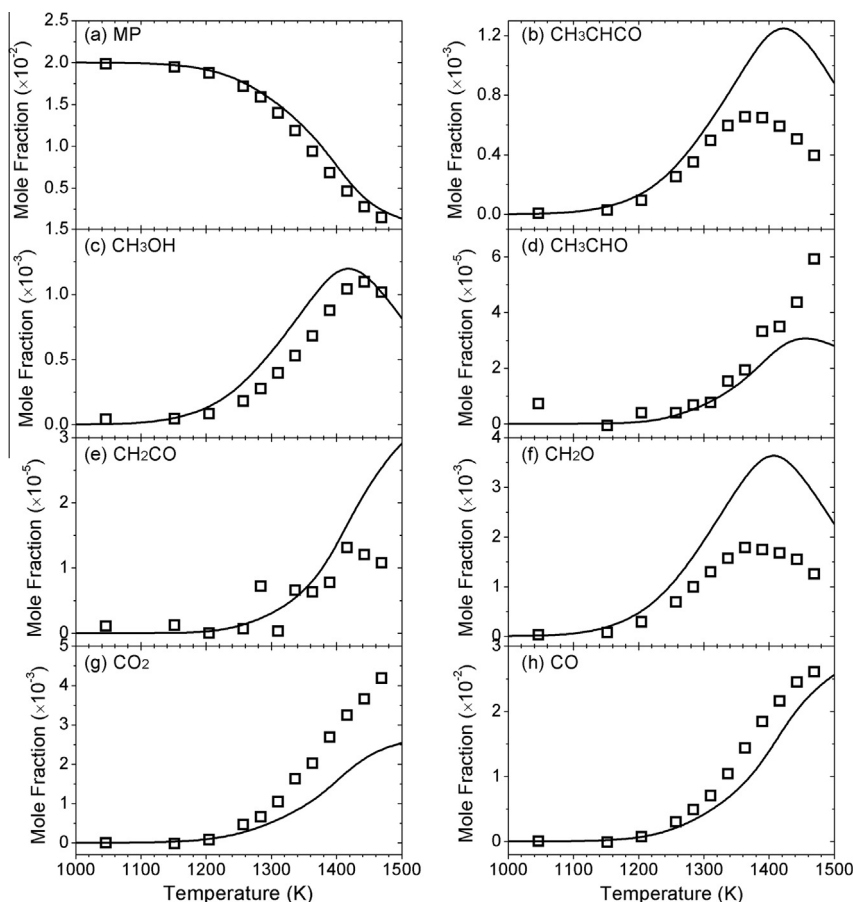


Fig. 4. Experimental (open squares) and simulated (lines) mole fraction profiles of MP and oxygenated products from MP pyrolysis.

pyrolysis of MP based on the ROP analysis. Therefore both CO and CO₂ have quite high mole fractions (10^{-2} – 10^{-3}), as shown in Fig. 4e and f.

The CH₃OH elimination reaction R3 (MP = CH₃OH + CH₃CHCO) contributes to 18% of MP consumption. In this work, a new channel of R3 is calculated, as shown in Fig. 2. Thus the reaction rates of R3 is different from the one calculated by Akih-Kumgeh and Bergthorson [7]. As seen in Fig. 4b and c, the mole fractions of CH₃CHCO and CH₃OH reach relatively high magnitudes of 10^{-3} , which indicates that they are major oxygenated intermediates in the pyrolysis of MP. This is also approved by Dievart et al. [1], who stated that the methanol elimination reaction has a certain tendency in the combustion of methyl esters smaller than methyl butanoate. The appearance temperatures of CH₃CHCO and CH₃OH are both near 1100 K, it is the same with the initial decomposition temperature of MP, which is consistent with the fact that the overall barriers of the channel R3 are the lowest (Fig. 2). The ROP analysis also exhibits that CH₃OH (~100%) and CH₃CHCO (85%) are dominantly formed by R3. Besides, the sensitivity analysis performed at 1400 K shows that R3 exhibits a much higher positive sensitivity than other reactions to the formation of CH₃OH and CH₃CHCO, as shown in Fig. 8.

CH₃OH is mainly consumed by H-abstraction reaction via H attack to produce of CH₂OH and CH₃O radicals (CH₃OH + H = CH₂OH + H₂, CH₃OH + H = CH₃O + H₂), which consume 74% and 17% of CH₃OH, respectively. CH₃O radical can also be formed by β -scission pathways of MP2J and CH₃OCO radicals. The latter radical is mainly formed from the unimolecular decomposition reaction of MP via the reaction MP = CH₃OCO + C₂H₅ (86%).

H-assisted cleavage (CH₃CHCO + H = C₂H₅ + CO) is the major consumption pathway for CH₃CHCO, contributing over half for CH₃CHCO decomposition.

H-abstraction by the H atom costs 29% CH₃CHCO to yield CH₂CHCO radical, which further completely breaks to C₂H₃ and CO. And the sequence (CH₃CHCO → CH₂CHCO → CO) has a contribution of 2.6% for CO formation.

As seen in Fig. 4d and e, CH₃CHO and CH₂CO are two oxygenated species detected with the peak mole fractions at the magnitude of 10^{-5} . The two species are not major products in MP pyrolysis, thus they are not described in detail here. They are both by-products from CH₂O decomposition pathways.

H-abstraction reactions of MP via H and CH₃ attack (R11–R16) totally contribute 19% consumption of MP, producing three C₄H₇O₂ radicals: CH₂CH₂COOCH₃ (MP3J), CH₃CHCOOCH₃ (MP2J), and CH₃CH₂COOCH₂ (MPMJ). Among these H-abstraction reactions, the abstraction of α -C–H in the alkyl chain has an apparent tendency, which consumes 16% of the fuel. The product MP2J has a resonantly stabilized structure which promotes this pathway. Similarly, the H atoms in the CH₃O group can also be abstracted easily, which has 2% contribution to the fuel decomposition. MP3J and MPMJ mainly decompose via β -scission reactions to produce C₂H₄ + CH₃OCO (R20) and CH₂O + C₂H₅CO (R21), respectively. Moreover, the decomposition reaction sequences of MPMJ contribute to production of CO via both C₂H₅CO and CH₂O. Most of MP2J (80%) is consumed through the isomerization reaction R9 to form MPMJ, while the rest suffers the β -scission reaction R19 to produce CH₃O + CH₃CHCO.

4.2. Formation of small hydrocarbons

As listed in Table 3, a number of hydrocarbon products were detected. For the aim to study the decomposition mechanism of MP, experimental and simulated mole fraction profiles of C1–C3

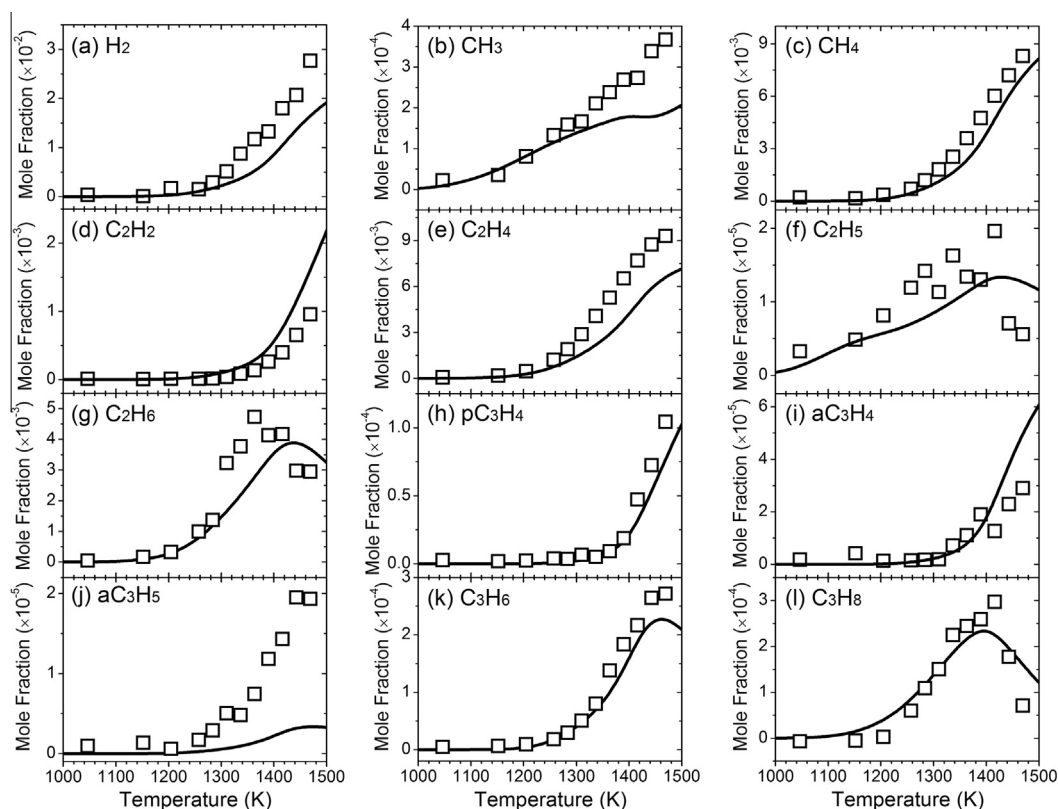


Fig. 5. Experimental (open squares) and simulated (lines) mole fraction profiles of C0–C3 hydrocarbons from MP pyrolysis.

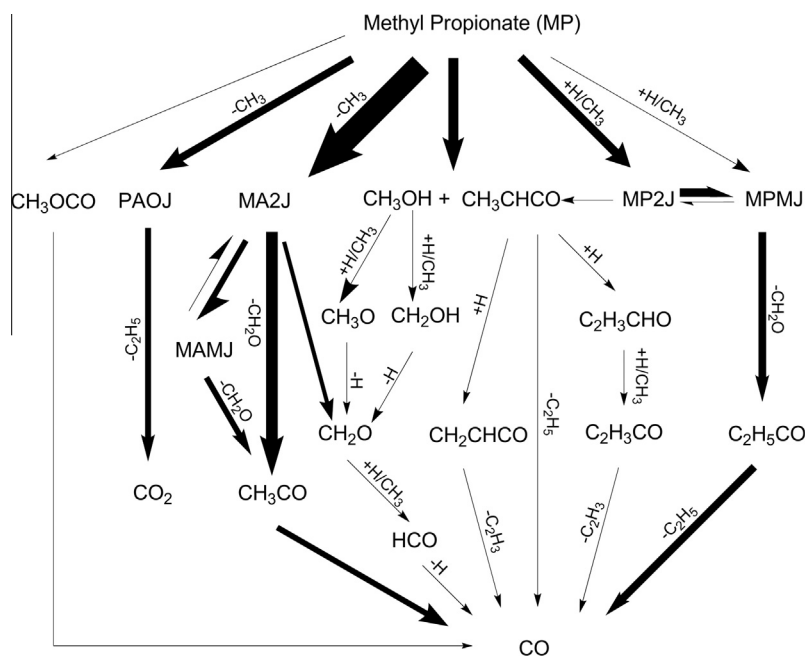


Fig. 6. Decomposition pathways of MP at 1400 K. The thicknesses of arrows reflect the fluxes of corresponding reactions.

hydrocarbon products and H_2 are shown in Fig. 5. As recombination products of small hydrocarbons, a few C4–C5 hydrocarbons were observed with quite low concentrations within the investigated temperature region, which implies that their formation pathways afford negligible reaction flux compared with those of C1–C3

hydrocarbons. Therefore the C4–C5 hydrocarbon products will not be discussed here. For convenience, reaction pathways of C0–C1 species are first discussed here.

Figure 5a–c shows the experimental and simulated mole fraction profiles of H_2 , CH_3 and CH_4 . The formation pathways of CH_4

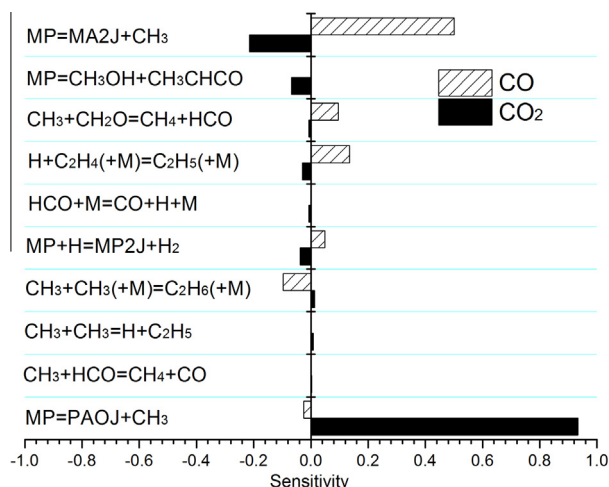


Fig. 7. Sensitivity analysis of CO and CO₂ from MP pyrolysis at the temperature of 1400 K. Only reactions with the top 10 sensitivity coefficients are considered.

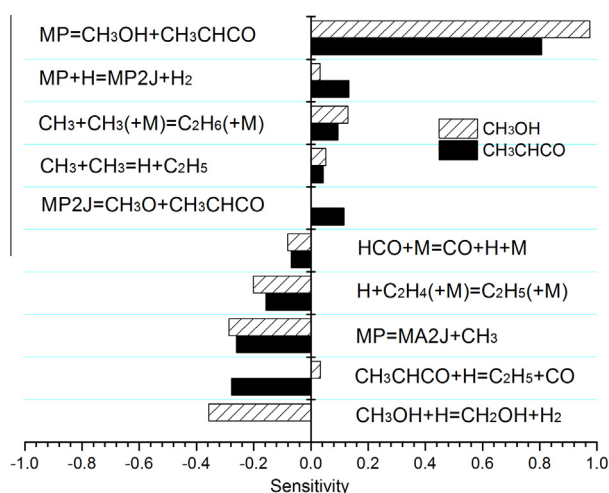


Fig. 8. Sensitivity analysis of CH₃OH and CH₃CHCO from MP pyrolysis at the temperature of 1400 K. Only reactions with the top 10 sensitivity coefficients are considered.

and H₂ are similar, both due to the H-abstraction reactions of larger species via the CH₃ and H attack, respectively. CH₄ and H₂ are mainly formed via H-abstraction reactions of CH₂O by CH₃ and H, with contribution of 55% (CH₂O + CH₃ = CH₄ + HCO) and 46% (CH₂O + H = H₂ + HCO), respectively.

Figure 5d–g presents the experimental and simulated mole fraction profiles of C₂ hydrocarbons, including C₂H₂, C₂H₄, C₂H₅ and C₂H₆. The ROP analysis shows that C₂H₆ is totally formed by the self-combination of CH₃ radical. Though C₂H₆ can lose H atom to produce C₂H₅ radical, the main precursors of C₂H₅ are primary and secondary decomposition products of MP such as C₂H₅CO (36%, C₂H₅CO = C₂H₅ + CO) and PAOJ (31%, PAOJ = C₂H₅ + CO₂). The β-scission of C₂H₅ is the dominant formation pathway of C₂H₄. The latter species can suffer sequential H-loss reactions to produce C₂H₂.

Figure 5h–l illustrates the experimental and simulated mole fraction profiles of C₃ hydrocarbons. Their maximum mole fractions are all among the magnitudes of 10^{−4}–10^{−5}. With the assistance of ROP analysis, it can be seen C₃H₈ is entirely formed via combination reaction of CH₃ and C₂H₅. C₃H₈ is consumed by H-abstraction reactions to form nC₃H₇ (48%) and iC₃H₇ (46%) radicals,

while both of the C₃H₇ radicals can produce C₃H₆ by β-C–H scission reactions. It is concluded that C₃H₈ and C₃H₆ are mainly formed from the combination of C₁ and C₂ hydrocarbons, and their decomposition lead to the formation of other C₃ hydrocarbons including aC₃H₅ and C₃H₄ isomers.

5. Conclusion

The experimental study of MP pyrolysis from 1000 to 1500 K was performed in a laminar flow reactor using SVUV-PIMS. Around 30 pyrolysis species were detected, and their mole fractions versus temperature were evaluated. The sub-mechanism of MP including unimolecular decomposition and H-abstraction reactions, and subsequent reactions of initial products are significant in MP pyrolysis. Thus the rate constants of these reactions were calculated in the present work to enhance the accuracy of the model. Two channels for the formation of CH₃CHCO and CH₃OH were found. One is the direct decomposition via a four-member-ring transition state. The other includes three steps, which are H-migration, H-rotation and CH₃OH-elimination. Based on the theoretical calculations, a detailed kinetic model of MP pyrolysis consisting of 98 species and 493 reactions was developed to simulate the pyrolysis experiment. The present model reproduces most of the measured mole fraction profiles of pyrolysis species. The simulated results of MP and other pyrolysis species have exclusively high sensitivities to the calculated rate constants of the primary decomposition reactions. The ROP analysis indicates that three unimolecular dissociation reactions (MP = CH₃ + CH₂COOCH₃, MP = CH₃ + CH₃CH₂COO, MP = CH₃CHCO + CH₃OH) are dominant consumption pathways of MP, which finally lead to the formation of CO and CO₂.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.combustflame.2013.04.022>.

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